SMS-STYLE PERSONALIZED, MULTIMEDIA NEWS EXPERIENCE

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Exploring Computational Journalism – Week 4, Phase 1.0 Demo

THE TEAM

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Ananya N. – *CS Undergrad*

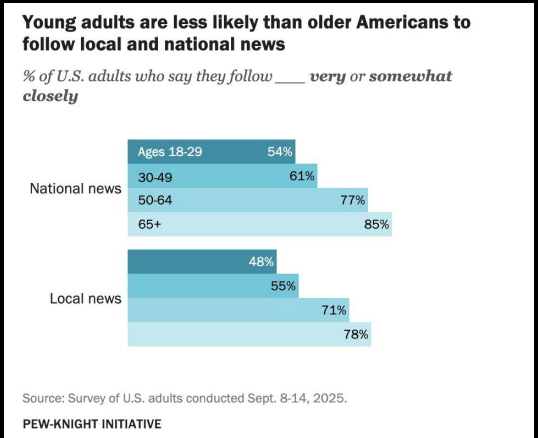
James V. – *CS Coterm*

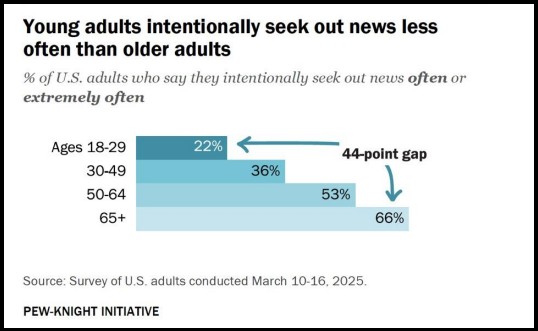
THE PROBLEM.

PROBLEM #1

NEWS HABITS ARE CHANGING.

- Young Americans **less likely to "seek out"** news (Pew Research, 2025)
- **54%** of Americans surveyed use **social media** for news (Reuters, 2025)
- A.I. chatbots are becoming **a source** for news
  - **7%** U.S. survey respondents consult chatbot for weekly news. This number goes up to **15%** of under-25s (Nieman Lab, 2026, World Economic Forum)





PROBLEM #2

PERSONALIZATION DEMANDS MORE CONTENT, BUT... DIFFERENT NEWSROOMS, DIFFERENT OBJECTIVES (& FORMATS)

KQED

Retain current users.

- Live stream listeners want ways to search the radio archives
- Not enough geography/region-specific content
- Chat-based conversational tool higher in demand

embarcadero media foundation

Draw younger audiences.

- More limited resources, journalists are less certain of A.I.
- No audio content & purely web-based
- Unsure of feasibility with current users (40+ years)

A SHARED FOCUS:

DEVELOP A LOYAL AUDIENCE & OPTIMIZE RETENTION.

THE PROBLEM.

OUR GOAL.

RECONNECT AUDIENCES WITH CONTENT THEY VALUE.

1. **OPTIMIZE SEARCH –** Enhance searchability of radio/audio feeds (KQED)

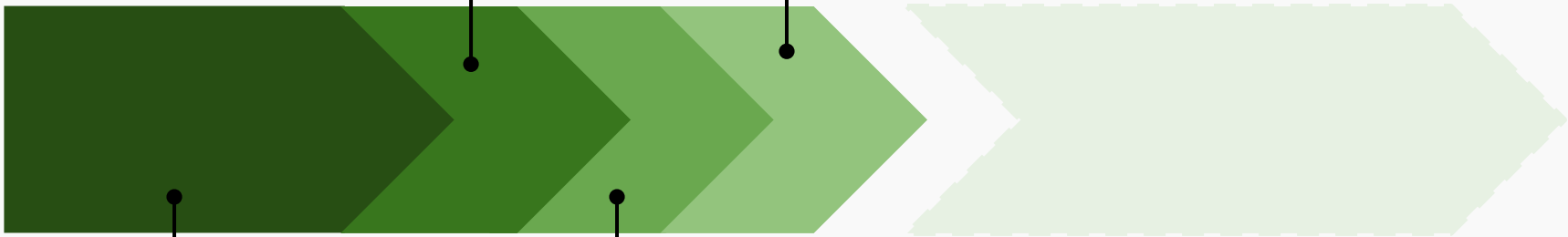
2. **COMBINE ARCHIVES TO OPTIMIZE FOR PERSONALIZATION –** Allow users to see more personalized content based on geography

3. **TEXT TO SPEECH/SPEECH TO TEXT –** Offer multimodal variety given newsroom content (audio, text, images, etc.)

UP 'TIL NOW...

1 Preliminary interviews with stakeholders/collaborating newsrooms 👍

3 Embarcadero RSS embedding, search & text-to-speech low-fi prototype 👍



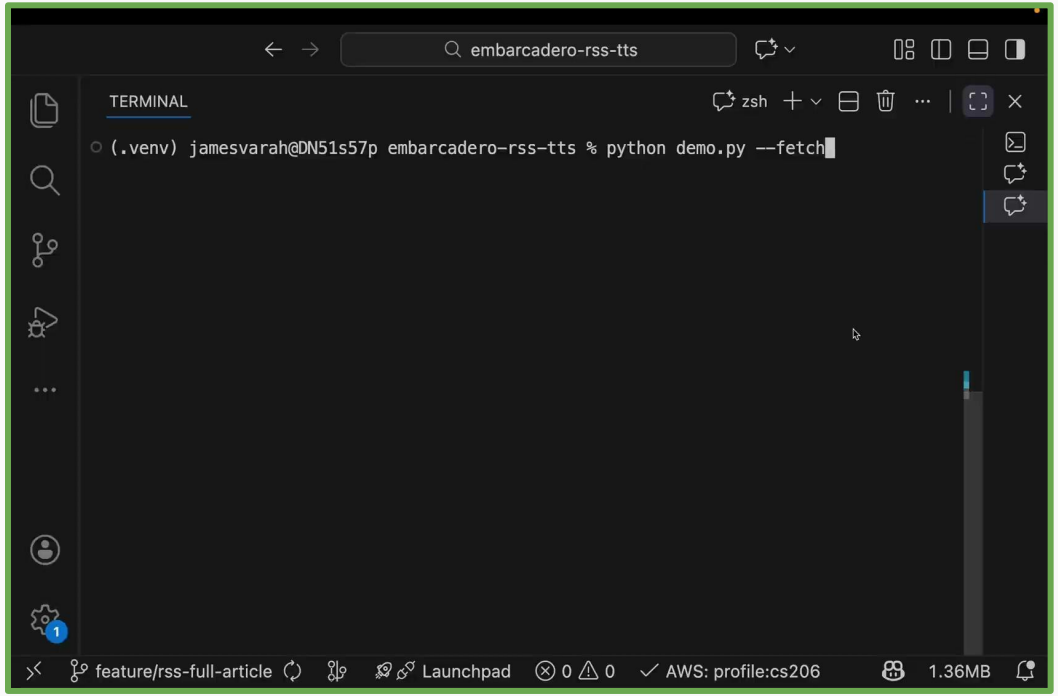
2 KQED live stream transcription, embedding & search low-fi prototype 👍

Team formed, project assigned & expectations set 👍

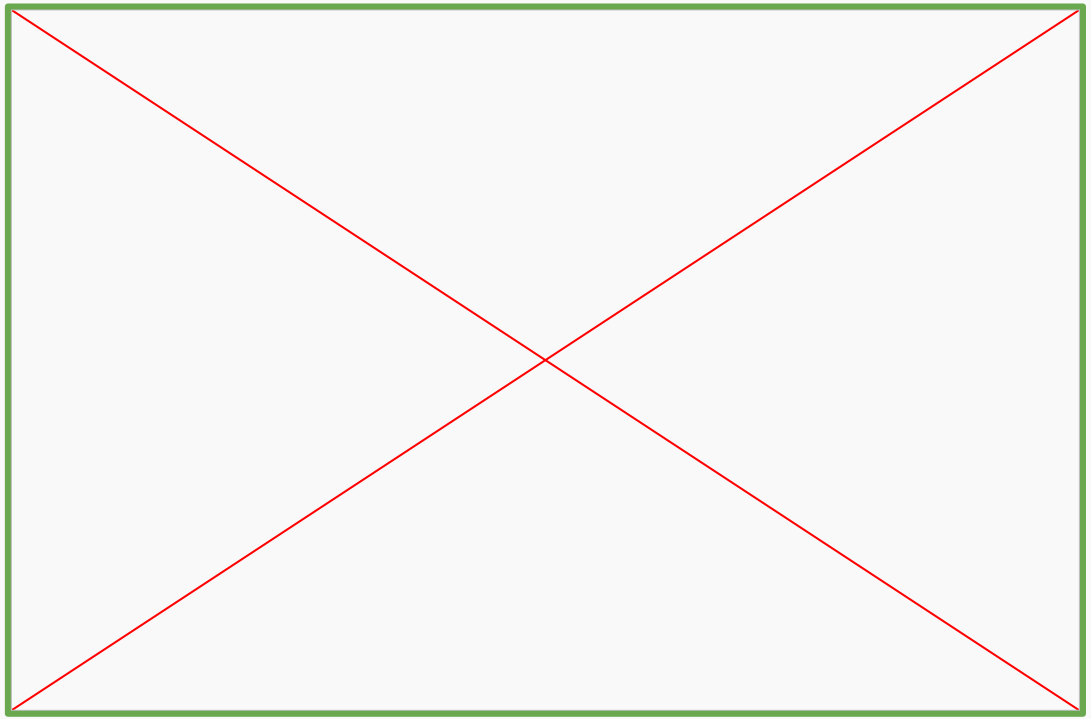
DEMO.

WOULD WE LISTEN TO TEXT-TO-SPEECH OF NEWS?

- **Audio** can be more engaging
- But **text** articles cover **local news** missed by radio
- Answer: **No**, it's not engaging. Leave text as text



CAN WE SEARCH THE RADIO ARCHIVE?



- **Yes**, as a user and as a curator
- Already informative and funny!

THE PROBLEM.

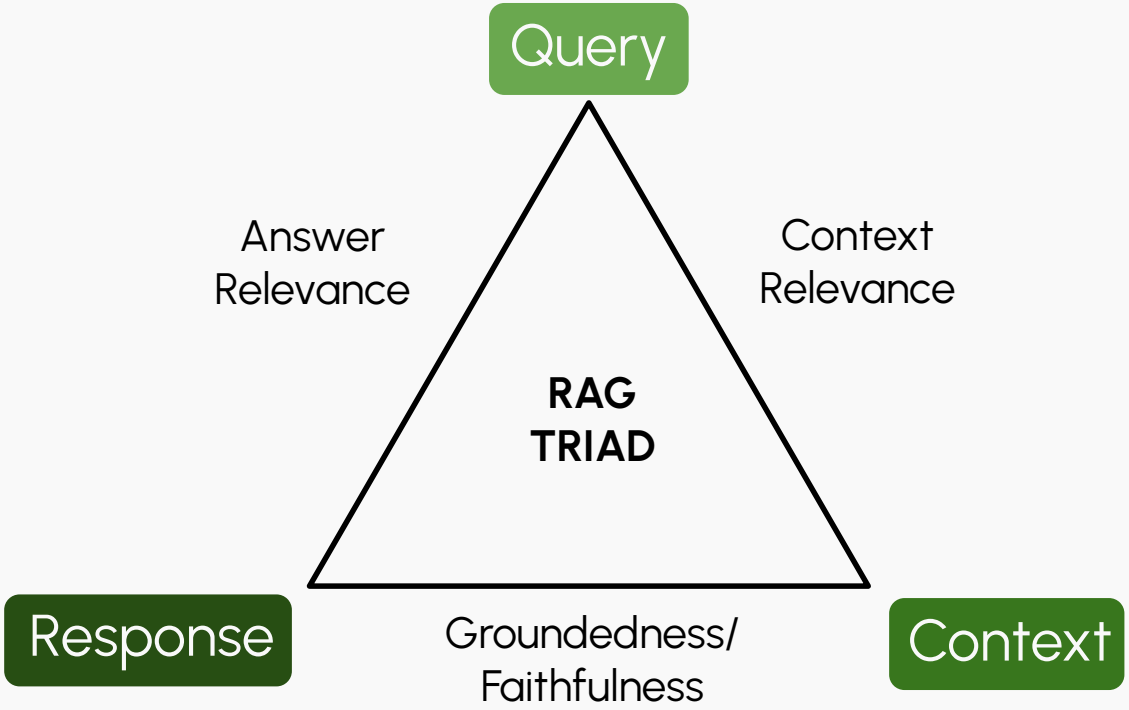
OUR GOAL.

NEXT STEPS.

ON TO THE NEXT!

- **Combine newsrooms content into one database**
- Iterate on current prototype for audio search
- **Incorporate accuracy-checking models**
- **Determine feasibility for geography**
- Develop simple UI for chatbot archive
- Interviews with journalists + potential users
- Pick a name → potentially… "Radio Flyer"?

ACCURACY METRICS



THINGS TO BE CAREFUL OF:

- Audio clips are relevant & clipped correctly
- Permitting the LLM to return "I don't know" or "this is my best guess" depending on p-value
- Human-in-the-loop accuracy of creating a "podcast" summary of an article

EXPLORING COMPUTATIONAL JOURNALISM

1. We look at ways to use computing to **distribute** news through:
    a. Aggregating local and regional news of **text** and **audio**
    b. Improving audio and text search for user-controlled personalization
2. Personalization will help with **funding** news by:
    a. Increasing traffic to local news organizations
    b. Improving user retention by showing **relevant** text articles and audio stories

THANK YOU.

Questions?